Nirma University

Institute of Technology

Department of Computer Science & Engineering

Explainable AI

Lab 1: Explainability in Intrinsically Explainable Models

(Regression, Decision Trees, Random Forest + SHAP)

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1 Objective

The objective of this lab is to demonstrate explainability in **intrinsically interpretable** machine learning models. Three models are explored:

1. **Multiple Linear Regression** — interpreting coefficients as feature importance
2. **Decision Tree Classifier** — visualizing tree structure and Gini importance
3. **Random Forest Classifier with SHAP** — using SHapley Additive exPlanations for post-hoc explainability

2 Datasets

2.1 Cellphone Dataset

- **Task:** Regression (predicting cellphone price)
- **Samples:** 161 entries
- **Features (12):** Sale, weight, resolution, ppi, cpu core, cpu freq, internal mem, ram, RearCam, Front_Cam, battery, thickness
- **Target:** Price (integer)
- **Preprocessing:** Standard scaling (zero mean, unit variance) applied to all features

2.2 MBA Admissions Dataset

- **Task:** Classification (predicting admission decision)
- **Samples:** 1,000 (with valid admission labels out of 6,194 total)
- **Features (34 after encoding):** gpa, gmat, work_exp, gender, international, major, race, work_industry (one-hot encoded)
- **Target:** Admission decision (categorical)
- **Preprocessing:** Rows with missing `admission` dropped; categorical features one-hot encoded

2.3 Wine Quality Dataset

- **Task:** Multi-class classification (predicting wine quality score)
- **Samples:** 1,599 entries
- **Features (11):** Fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free/total sulfur dioxide, density, pH, sulphates, alcohol
- **Target:** Quality (integer score)
- **Split:** 80/20 train-test split

3 Methodology

3.1 Part 1: Multiple Linear Regression

3.1.1 Model Description

Multiple Linear Regression models the target variable as a weighted linear combination of input features:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p \tag{1}$$

where β_0 is the intercept and $\beta_1, \ldots, \beta_p$ are the regression coefficients. Each coefficient β_j quantifies the expected change in the target for a one-unit change in feature x_j, holding all other features constant.

3.1.2 Implementation Steps

1. Loaded `Cellphone.csv` and inspected dataset structure using `df.info()`
2. Separated the target variable (`Price`) from features; dropped `Product_id`
3. Applied `StandardScaler` to normalize all features to zero mean and unit variance
4. Fitted `LinearRegression` from scikit-learn on the entire dataset
5. Extracted model coefficients (`model.coef_`) for explainability analysis

3.1.3 Explainability Approach

Since features are standardized, the **absolute magnitude of coefficients** serves as a direct measure of feature importance. A larger absolute coefficient indicates a stronger influence on the predicted price.

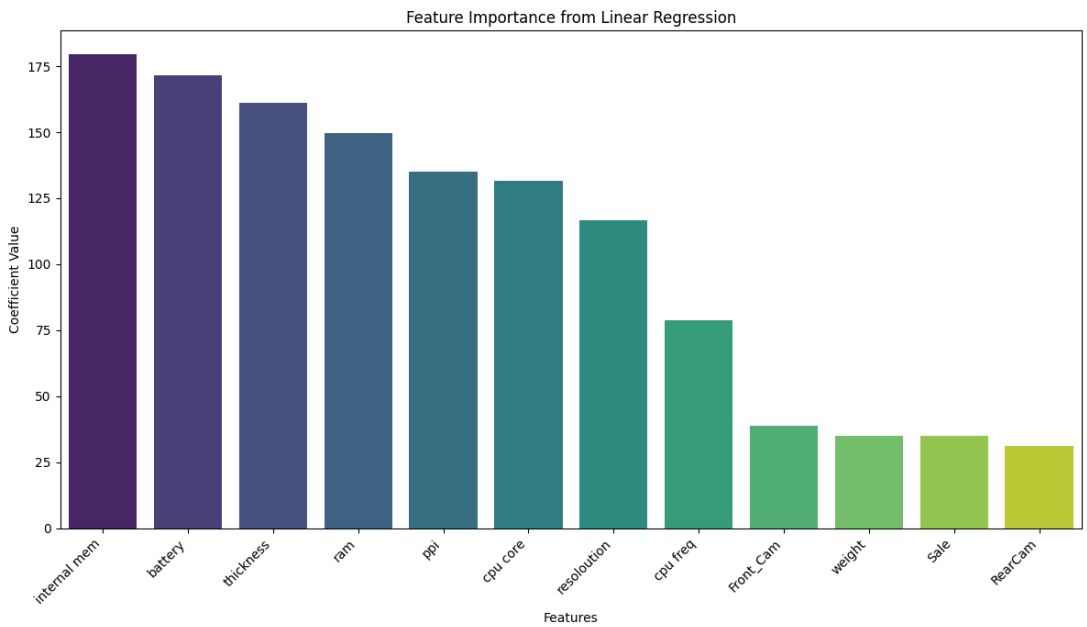

Figure 1: Feature importance derived from Linear Regression coefficients (absolute values). Internal memory, battery, and thickness show the highest importance.

3.1.4 Results Interpretation

Key coefficients obtained:

- `internal mem`: +179.47 — strongest positive predictor of price
- `battery`: +171.56 — second strongest positive predictor
- `thickness`: −161.06 — strong negative effect (thinner phones cost more)
- `ram`: +149.57 and `ppi`: +135.18 — significant positive contributors
- `Sale`: −35.07 and `weight`: −35.15 — minor negative effects

The coefficient signs and magnitudes provide a complete, transparent explanation of the model's pricing logic.

3.2 Part 2: Decision Tree Classifier

3.2.1 Model Description

A Decision Tree recursively partitions the feature space using axis-aligned splits that maximize class purity. At each internal node, the algorithm selects the feature and threshold that best separates the classes (using Gini impurity in this case):

$$\text{Gini}(t) = 1 - \sum_{k=1}^{K} p_k^2 \tag{2}$$

where p_k is the proportion of class k at node t.

3.2.2 Implementation Steps

1. Loaded `MBA.csv` and filtered to 1,000 rows with valid admission labels
2. One-hot encoded categorical features (gender, international, major, race, work_industry), yielding 34 features
3. Trained `DecisionTreeClassifier` with `random_state=42`
4. Extracted `feature_importances_` (mean decrease in Gini impurity)
5. Visualized the complete tree structure using `plot_tree`

3.2.3 Explainability Approach

Decision trees are inherently interpretable because:

- Every prediction can be traced through a sequence of if-then rules
- Feature importance is quantified by the total reduction in Gini impurity contributed by each feature across all splits
- The tree structure itself serves as the explanation

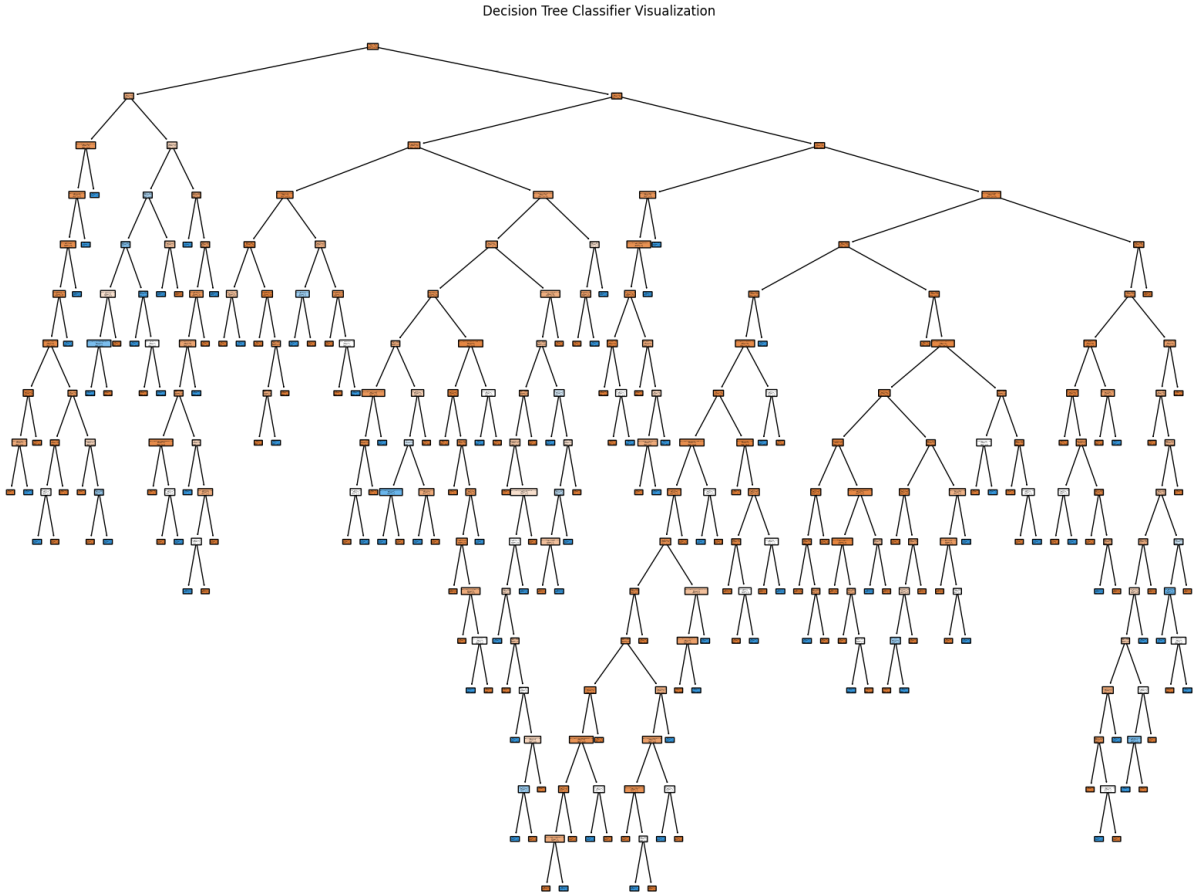

Figure 2: Full Decision Tree visualization for MBA admission prediction. Nodes are colored by majority class, with split conditions and sample counts shown.

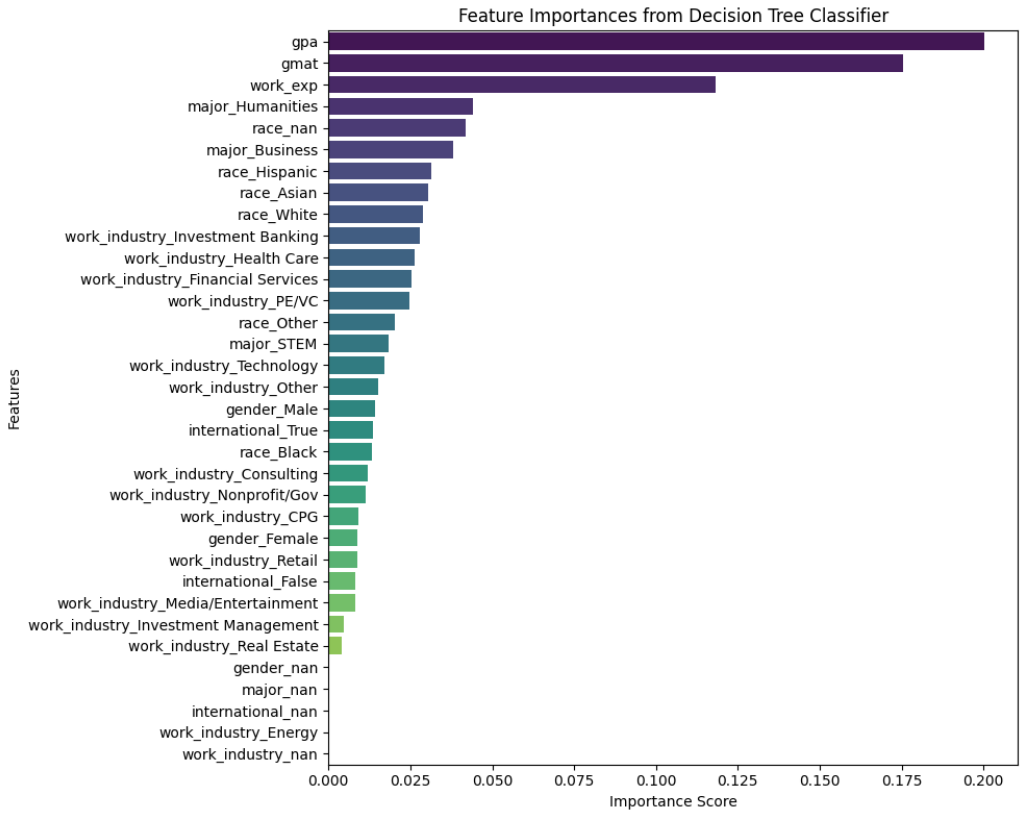

Figure 3: Feature importance scores from Decision Tree. GPA, GMAT, and work experience dominate the decision process.

3.2.4 Results Interpretation

- **GPA (0.200)** is the single most important feature for admission
- **GMAT (0.175)** and **work experience (0.118)** follow closely
- Demographic features (gender, race, international status) have minimal importance (< 0.015), suggesting the model relies primarily on academic and professional merit

3.3 Part 3: Random Forest with SHAP

3.3.1 Model Description

A Random Forest is an ensemble of B decision trees, each trained on a bootstrap sample with random feature subsets. The final prediction is the majority vote:

$$\hat{y} = \text{mode}\{h_1(\mathbf{x}), h_2(\mathbf{x}), \ldots, h_B(\mathbf{x})\} \tag{3}$$

While individual trees are interpretable, the ensemble is a **black box**. SHAP (SHapley Additive exPlanations) provides post-hoc explainability.

3.3.2 SHAP Theory

SHAP values are based on Shapley values from cooperative game theory. For a prediction $f(\mathbf{x})$:

$$f(\mathbf{x}) = \phi_0 + \sum_{j=1}^{p} \phi_j \tag{4}$$

where ϕ_0 is the base value (average prediction) and ϕ_j is the SHAP value for feature j, representing its marginal contribution.

3.3.3 Implementation Steps

1. Loaded `winequality-red.csv` and split into train/test (80/20)
2. Trained `RandomForestClassifier` with default hyperparameters
3. Initialized `shap.TreeExplainer` for efficient SHAP computation on tree ensembles
4. Computed SHAP values for all test samples
5. Generated a SHAP summary plot

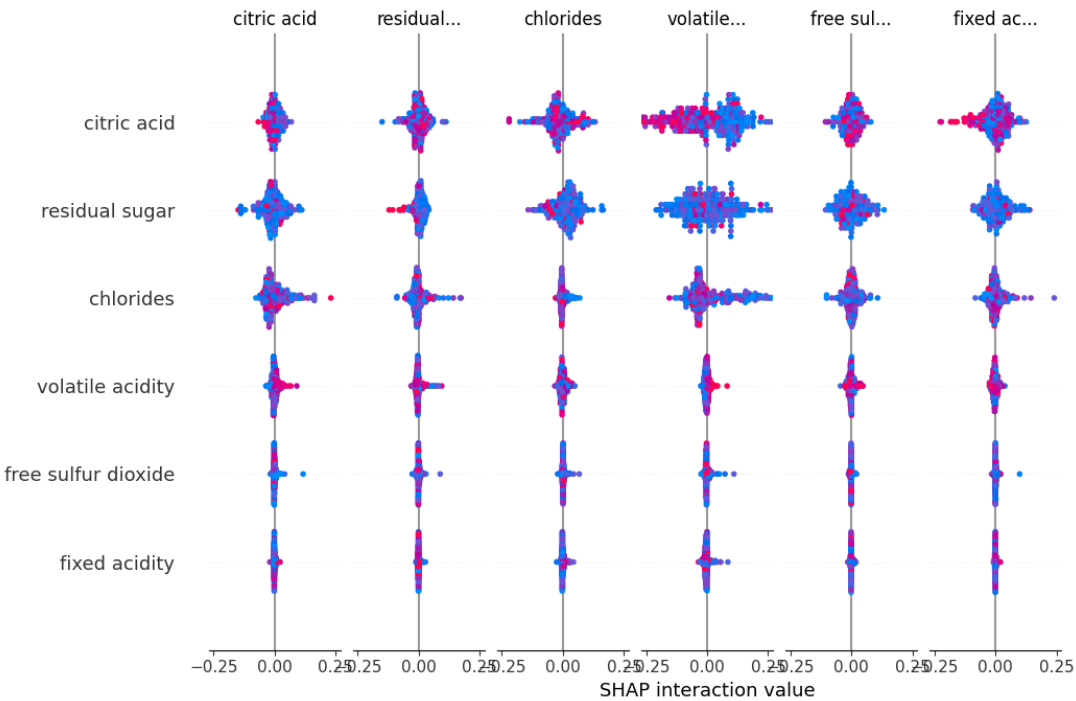

Figure 4: SHAP summary plot for Random Forest on wine quality. Each row is a feature; each dot is a test sample. Color indicates feature value (red=high, blue=low); x-axis shows SHAP value (impact on prediction).

3.3.4 Results Interpretation

- **Alcohol** has the widest spread of SHAP values, making it the most influential feature for wine quality

- **Volatile acidity** and **sulphates** are the next most important
- High alcohol content (red dots) consistently pushes predictions toward higher quality (positive SHAP values)
- High volatile acidity (red dots) pushes predictions toward lower quality (negative SHAP values)

4 Conclusion

This lab demonstrated that intrinsically explainable models offer transparent and interpretable predictions:

- **Linear Regression** provides direct coefficient-based feature importance
- **Decision Trees** offer rule-based explanations through their tree structure
- **SHAP** bridges the gap between complex models (Random Forest) and interpretability by providing theoretically grounded feature attributions

These methods form the foundation of Explainable AI, enabling practitioners to understand, trust, and debug their machine learning models.